

# CS 486/686

# Probabilities

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Lecture 6

RN 12.2–12.3 · PM 8.1

# Reminders

## DUE TONIGHT

Thu May 28, 11:59 PM

Chat 2 (Heuristic search & CSPs) on Chrysalis.

## OUT TODAY

Due Tue Jun 2

Chat 3 (Probabilities) — released today.

Search ›

Uncertainty ›

Decisions ›

Learning

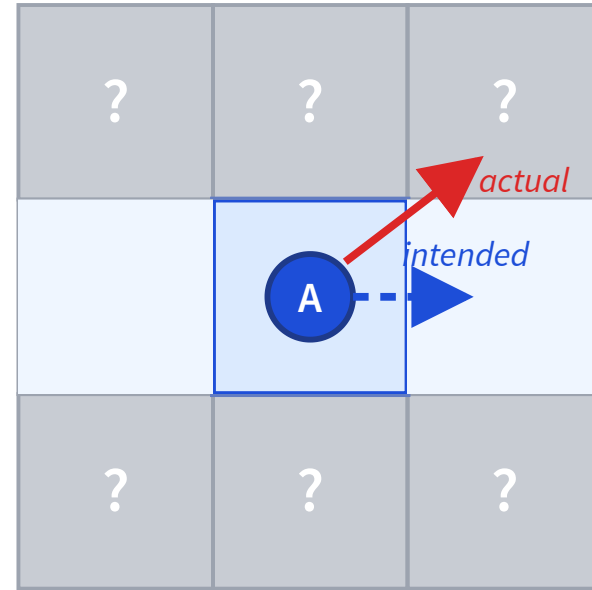
# Learning goals

Calculate prior, posterior, and joint probabilities using:

- the **sum rule** (marginalization),
- the **product rule** (conditioning),
- the **chain rule** (joint factorization),
- and **Bayes' rule** (flipping a conditional).
- Then put it all together with a **universal 3-step recipe**.

# Why handle uncertainty?

- Can't see the whole state — partial observation.
- Actions may not produce their intended outcome.
- **Reason about uncertainty** and decide anyway.



Visible cells are clear; the rest are hidden.  
Actions drift.

# Frequentists vs Bayesians

Two camps for the formal measure of uncertainty.

## Frequentist

Long-run frequency. *Objective*. After  $n$

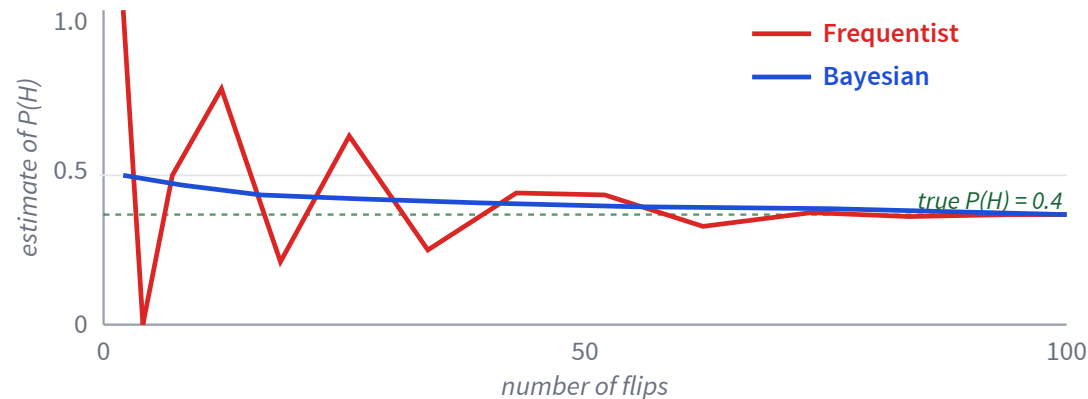
$$H, m T: P(H) = \frac{n}{n+m}. \text{ Needs data.}$$

## Bayesian

Degree of belief: prior + evidence.

$$\text{Laplace 1H/1T: } P(H) = \frac{1+n}{2+n+m}.$$

Works with no data.

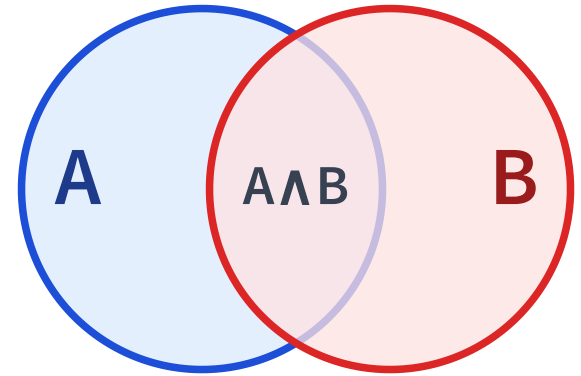


# Random variables

- A *domain* of values + a *distribution* over them.
- e.g.  $A$  = “alarm sounding”: domain  $\{T, F\}$ ,  $P(A=T) = 0.1$ ,  $P(A=F) = 0.9$ .  
.
- Boolean shorthand:  $P(A) \equiv P(A=T)$ ;  $P(\neg A) \equiv P(A=F)$ .

# Axioms of probability

- $0 \leq P(A) \leq 1$
- $P(\text{true}) = 1$ ;  $P(\text{false}) = 0$
- **Inclusion-exclusion:**  $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$



$0 < P(A) < 1$  means we're *uncertain*, not that  $A$  is partly true.

# Joint probability distribution

- **Atomic event:** an assignment to every variable.
- **Joint distribution:** a probability for every atomic event.

e.g.  $P(\text{weather, temperature})$

|        | Hot  | Mild | Cold |
|--------|------|------|------|
| Sunny  | 0.10 | 0.20 | 0.10 |
| Cloudy | 0.05 | 0.35 | 0.20 |

Probabilities sum to 1 across the table.



# Prior vs posterior

- Prior  $P(X)$  – belief in  $X$  with no other info.
- Posterior  $P(X \mid Y)$  – belief in  $X$  given evidence  $Y$ .



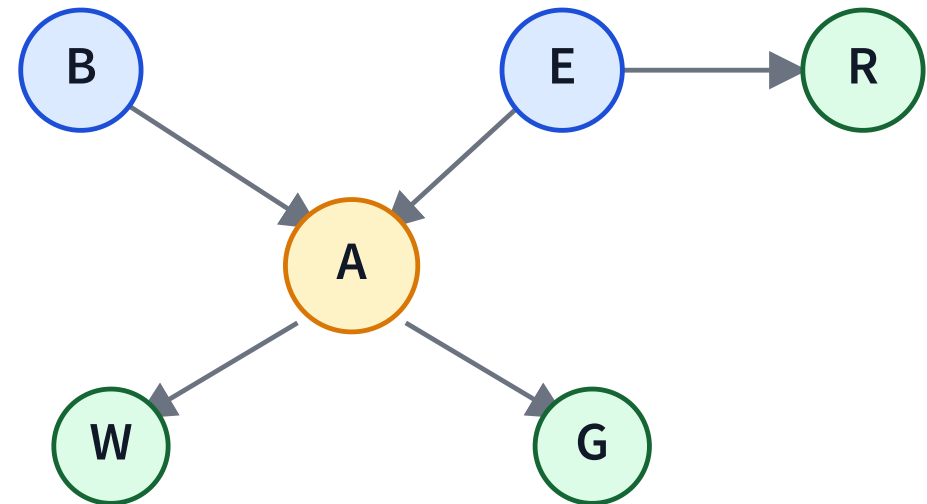
# Running example: Holmes' burglar alarm

Mr. Holmes has a burglar alarm and two flaky neighbours (Watson, Gibbon) who call when they hear it. The alarm also misfires on earthquakes; earthquakes are reported on the radio.

6 Boolean variables  $\Rightarrow 2^6 = 64$  joint probabilities.

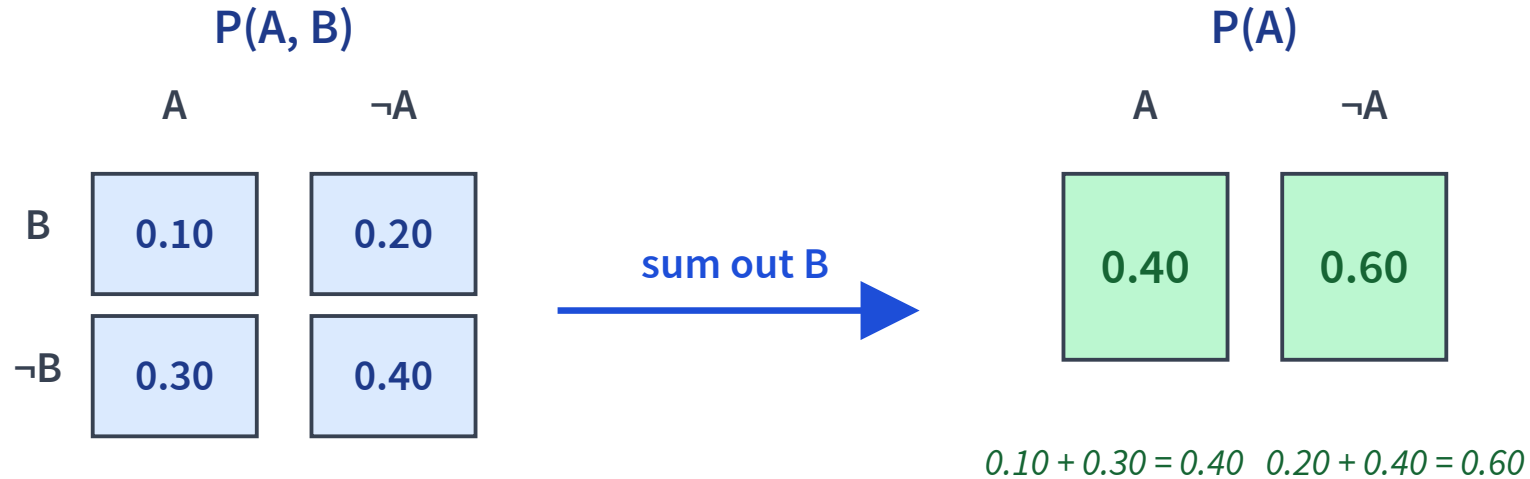
## Random variables

- $B$ : burglary
- $E$ : earthquake
- $A$ : alarm sounding
- $W$ : Watson calling
- $G$ : Gibbon calling



# The sum rule (marginalization)

From a joint, get a probability over a *subset* by summing out the rest.



In general:  $P(A) = \sum_b P(A \wedge B=b)$

# Marginalizing the Holmes joint

$P(A, W, G)$  over Watson  $W$ , Gibbon  $G$ , and the alarm  $A$ :

|          | $A$   |          | $\neg A$ |          |
|----------|-------|----------|----------|----------|
|          | $G$   | $\neg G$ | $G$      | $\neg G$ |
| $W$      | 0.032 | 0.048    | 0.036    | 0.324    |
| $\neg W$ | 0.008 | 0.012    | 0.054    | 0.486    |

Example —  $P(\neg A \wedge W)$ : sum out  $G$  within the highlighted cells.

$$P(\neg A \wedge W) = 0.036 + 0.324 = \mathbf{0.36}$$

# Quiz pack: marginalization

Q1.  $P(\neg A \wedge W)$ ?

$$= 0.036 + 0.324 = \mathbf{0.36}.$$

Q2.  $P(A \wedge \neg G)$ ?

$$= 0.048 + 0.012 = \mathbf{0.06}.$$

Q3.  $P(\neg A)$ ?

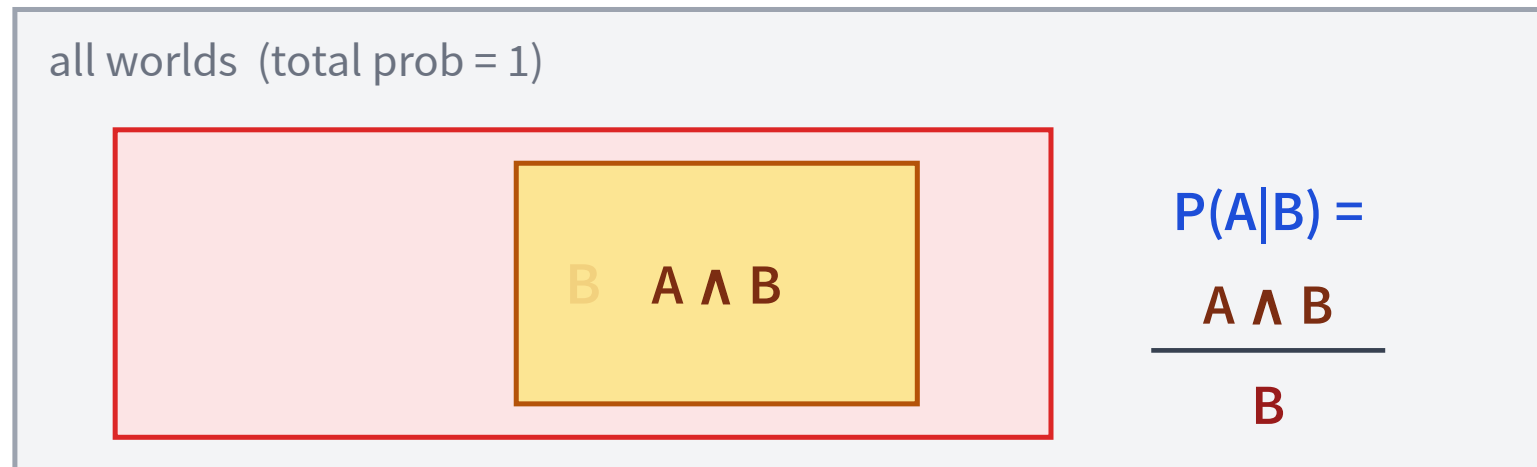
$$= 0.036 + 0.324 + 0.054 + 0.486 = \mathbf{0.9}.$$

|          | $A$   |          | $\neg A$ |          |
|----------|-------|----------|----------|----------|
|          | $G$   | $\neg G$ | $G$      | $\neg G$ |
| $W$      | 0.032 | 0.048    | 0.036    | 0.324    |
| $\neg W$ | 0.008 | 0.012    | 0.054    | 0.486    |

# Conditional probability and the product rule

$P(A|B)$  = fraction of the  $B$ -world where  $A$  also holds.

$$P(A|B) = \frac{P(A \wedge B)}{P(B)}$$



Multiply both sides by  $P(B)$ :

$$P(A \wedge B) = P(A|B) P(B)$$

$$\text{Q: } P(W \mid \neg A)?$$

From the previous slides:  $P(\neg A \wedge W) = 0.36$ ,  $P(\neg A) = 0.9$ .

Q. Watson calls, given the alarm is NOT going.

A. 0.2

B. 0.4

C. 0.6

D. 0.8

$$\text{B. } P(W \mid \neg A) = \frac{P(\neg A \wedge W)}{P(\neg A)} = \frac{0.36}{0.9} = \mathbf{0.4}.$$

$$\text{Q: } P(\neg G \mid A)?$$

From the previous slides:  $P(A \wedge \neg G) = 0.06$ ,  $P(\neg A) = 0.9$  (so  $P(A) = 0.1$ ).

Q. Gibbon does NOT call, given the alarm is going.

A. 0.2

B. 0.4

C. 0.6

D. 0.8

$$\text{c. } P(\neg G \mid A) = \frac{P(A \wedge \neg G)}{P(A)} = \frac{0.06}{0.1} = \mathbf{0.6}.$$



# The chain rule (joint factorization)

The product rule, repeated.

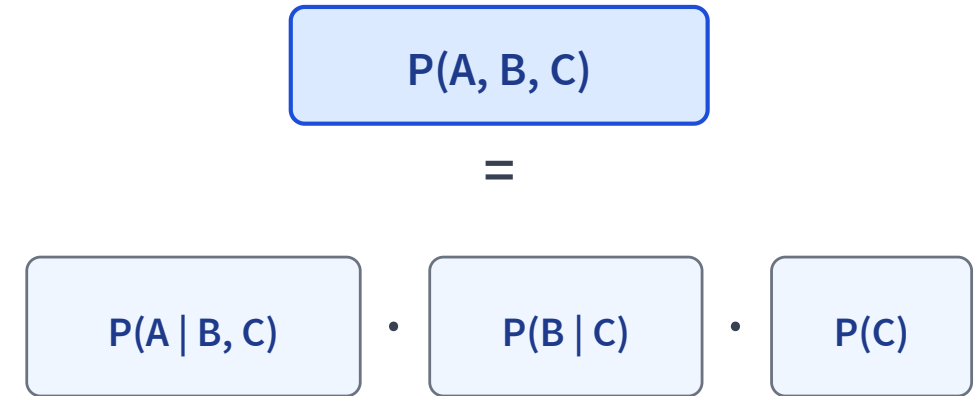
2 vars:  $P(A, B) = P(A|B) P(B)$

3 vars:

$$P(A, B, C) = P(A|B, C) P(B|C) P(C)$$

In general:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i | X_1, \dots, X_{i-1})$$



*condition each on its predecessors  
(any ordering works)*

$$\text{Q: } P(A \wedge W \wedge \neg G)?$$

Given:  $P(A) = 0.1$ ,  $P(W|A) = 0.9$ ,  $P(G|A \wedge W) = 0.3$ .

Q. Apply the chain rule to the conjunction.

A. 0.027

B. 0.037

C. 0.054

D. 0.063

$$\text{D. } P(A) \cdot P(W|A) \cdot P(\neg G|A \wedge W) = 0.1 \times 0.9 \times 0.7 = \mathbf{0.063}.$$

$$\mathbf{Q: } P(\neg A \wedge \neg W \wedge \neg G)?$$

Given:  $P(A) = 0.1$ ,  $P(W|\neg A) = 0.4$ ,  $P(G|\neg A \wedge \neg W) = 0.1$ .

**Q.** Apply the chain rule, this time to the all-negated conjunction.

A. 0.486

B. 0.324

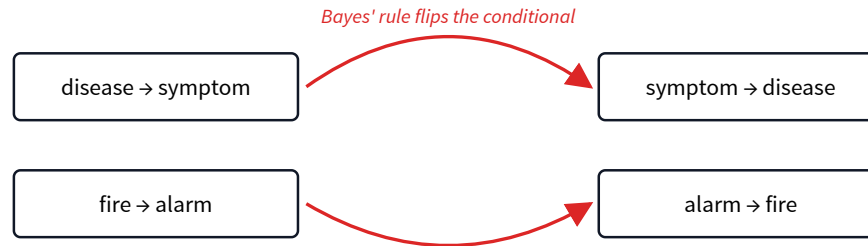
C. 0.216

D. 0.054

$$\mathbf{A. } P(\neg A) \cdot P(\neg W|\neg A) \cdot P(\neg G|\neg A \wedge \neg W) = 0.9 \times 0.6 \times 0.9 = \mathbf{0.486}.$$

# Why flip conditionals?

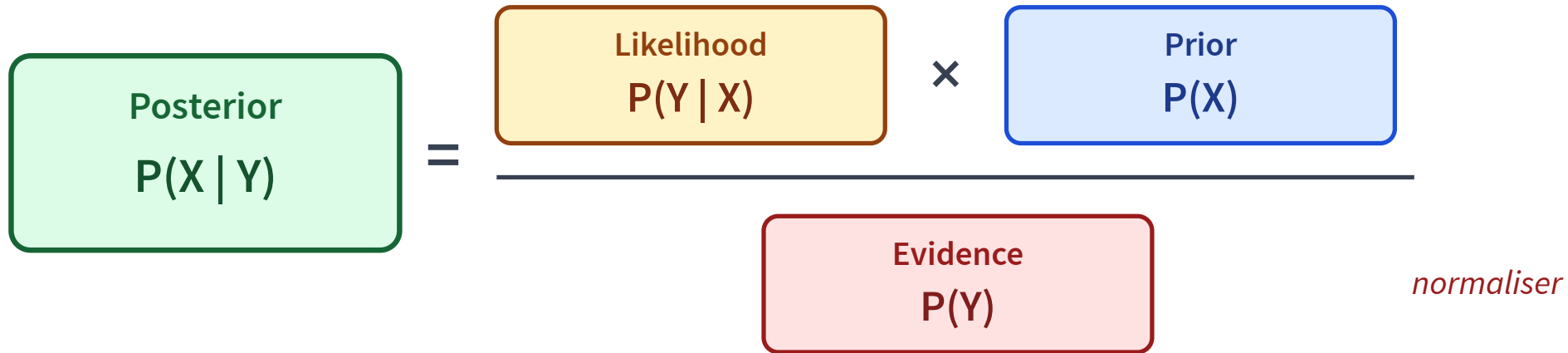
Models give us **causal** knowledge. Diagnosis needs **evidential** reasoning.



We know  $P(\text{symptom}|\text{disease})$  and want  $P(\text{disease}|\text{symptom})$ .

# Bayes' rule

Bayes' rule.  $P(X|Y) = \frac{P(Y|X) P(X)}{P(Y)}$



A diagram illustrating the components of Bayes' rule. On the left, a green rounded rectangle contains the text "Posterior" and  $P(X | Y)$ . To its right is an equals sign. Further right is a fraction. The numerator consists of two rounded rectangles: a yellow one labeled "Likelihood" and  $P(Y | X)$ , followed by a multiplication symbol  $\times$ , and then a blue one labeled "Prior" and  $P(X)$ . A horizontal line separates the numerator from the denominator. The denominator is a red rounded rectangle labeled "Evidence" and  $P(Y)$ . To the right of the denominator box, the word "normaliser" is written in red italicized font.

$$\text{Posterior } P(X | Y) = \frac{\text{Likelihood } P(Y | X) \times \text{Prior } P(X)}{\text{Evidence } P(Y)}$$

*normaliser*

Derive it from the product rule:  $P(X, Y) = P(X|Y) P(Y) = P(Y|X) P(X)$   
 $\rightarrow$  divide by  $P(Y)$ .

$$\text{Q: } P(\neg A \mid W)?$$

Given:  $P(A) = 0.1$ ,  $P(W|A) = 0.9$ ,  $P(W|\neg A) = 0.4$ .

Q. Alarm NOT going, given Watson calls.

- A. 0.2
- B. 0.45
- C. 0.8
- D. 0.9

C. Marginalize:  $P(W) = P(A)P(W|A) + P(\neg A)P(W|\neg A) = 0.09 + 0.36 = 0.45$ .

$$\text{Bayes: } P(\neg A|W) = \frac{P(\neg A)P(W|\neg A)}{P(W)} = \frac{0.36}{0.45} = \mathbf{0.8}.$$

$$\text{Q: } P(A \mid \neg G)?$$

Given:  $P(A) = 0.1, P(G|A) = 0.3, P(G|\neg A) = 0.1$ .

Q. Alarm going, given Gibbon does NOT call.

A. 0.03

B. 0.08

C. 0.30

D. 0.70

B. Marginalize:  $P(\neg G) = P(A)P(\neg G|A) + P(\neg A)P(\neg G|\neg A) = 0.07 + 0.81 = 0.88$ .

$$\text{Bayes: } P(A|\neg G) = \frac{P(A)P(\neg G|A)}{P(\neg G)} = \frac{0.07}{0.88} \approx \mathbf{0.08}.$$

# A universal approach

Three rules — one per kind of probability you need.

1. **To calculate a conditional probability** — write it as a ratio of two joint probabilities (*product rule, reversed*):  $P(X|Y) = \frac{P(X \wedge Y)}{P(Y)}$ .
2. **To calculate a partial joint** (some variables missing) — sum over the missing variables to get full joints (*sum rule, reversed*).
3. **To calculate a full joint** (all variables) — factor as a product of conditionals (*chain rule*).



# Worked example: $P(A|C)$

Given:

$$P(A) = 0.6$$

$$P(B|A) = 0.4, P(\neg B|\neg A) = 0.2$$

$$P(C|A, B) = 0.1, P(C|\neg A, B) = 0.2, P(C|A, \neg B) = 0.5, P(C|\neg A, \neg B) = 0.8$$

**Step 1.** Convert the conditional into a ratio of joints:

$$P(A|C) = \frac{P(A \wedge C)}{P(C)} = \frac{P(A \wedge C)}{P(A \wedge C) + P(\neg A \wedge C)}$$

## Worked example: step 2 (marginalize)

From step 1:  $P(A|C) = \frac{P(A \wedge C)}{P(A \wedge C) + P(\neg A \wedge C)}$ . The numerator and denominator are partial joints — sum out the missing variable  $B$ .

**Step 2.** Sum out  $B$ :

$$\begin{aligned}P(A \wedge C) &= P(A \wedge B \wedge C) + P(A \wedge \neg B \wedge C) \\P(\neg A \wedge C) &= P(\neg A \wedge B \wedge C) + P(\neg A \wedge \neg B \wedge C)\end{aligned}$$

Now every term on the right is a full-joint probability, ready for the chain rule.

## Worked example: step 3 + answer

Apply the chain rule  $P(A \wedge B \wedge C) = P(C|A \wedge B) P(B|A) P(A)$  to each full joint.

$$P(A \wedge B \wedge C) = 0.1 \times 0.4 \times 0.6 = 0.024$$

$$P(A \wedge \neg B \wedge C) = 0.5 \times 0.6 \times 0.6 = 0.180$$

$$P(\neg A \wedge B \wedge C) = 0.2 \times 0.8 \times 0.4 = 0.064$$

$$P(\neg A \wedge \neg B \wedge C) = 0.8 \times 0.2 \times 0.4 = 0.064$$

$$P(A|C) = \frac{0.024 + 0.180}{0.024 + 0.180 + 0.064 + 0.064} = \frac{0.204}{0.332} \approx \mathbf{0.614}$$

# Four rules at a glance

| Rule    | Formula                                                                             | Use when                                 |
|---------|-------------------------------------------------------------------------------------|------------------------------------------|
| Sum     | $P(A) = \sum_b P(A \wedge B=b)$                                                     | marginalizing out variables              |
| Product | $P(A \wedge B) = P(A B) P(B)$                                                       | defining or applying a conditional       |
| Chain   | $P(X_1 \wedge \dots \wedge X_n) = \prod_i P(X_i   X_1 \wedge \dots \wedge X_{i-1})$ | computing a full joint from conditionals |
| Bayes'  | $P(X Y) = \frac{P(Y X) P(X)}{P(Y)}$                                                 | flipping causal $\rightarrow$ evidential |

# Learning goals (recap)

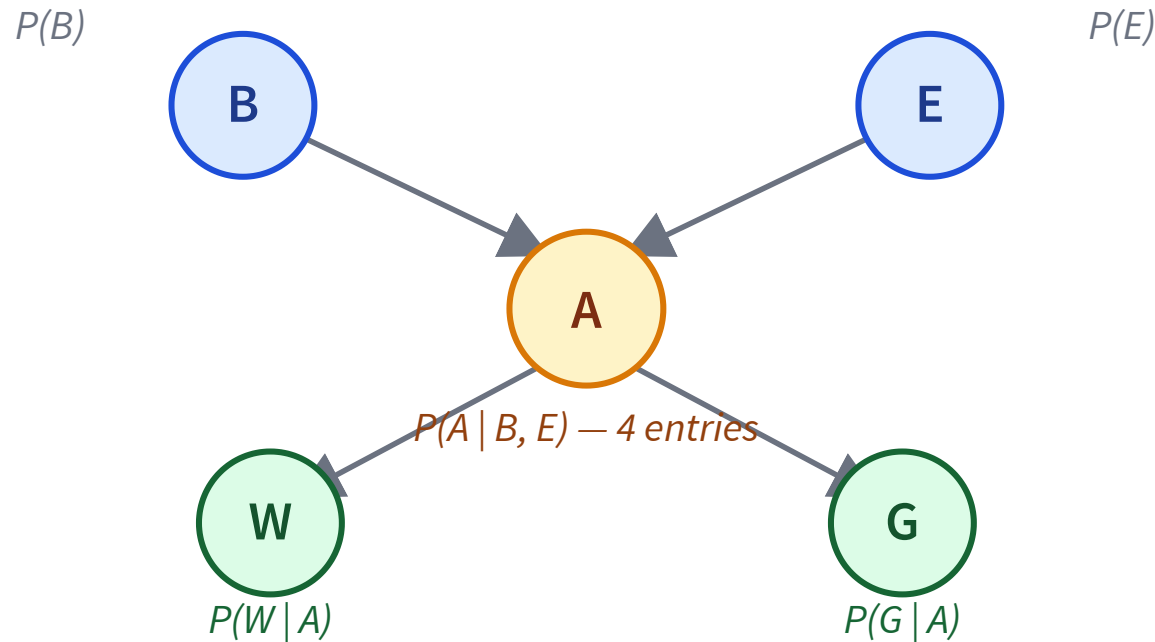
Calculate prior, posterior, and joint probabilities using:

- ✓ the **sum rule**
- ✓ the **product rule**
- ✓ the **chain rule**
- ✓ **Bayes' rule**
- ✓ the **universal 3-step recipe**

# Next: independence and Bayes nets

Chain rule gives full joints — but  $2^n$  numbers is too many.

**Conditional independence + Bayesian networks** → far fewer numbers.



*5 variables: 32 joint numbers → ~10 CPT entries.*